

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER – II) END TERM EXAMINATION, JANUARY-2025****Operations Management****[Time: 1:00 Hours]****[Maximum Marks: 30]****Following are the course outcomes of the subject: - Operation Management**

CO	Course Outcome (CO)
CO1	Understand the fundamental, practical science of Operations Management
CO2	Examine how these principles operations management can be employed in both tactical and strategic decision making in firms.
CO3	Develop ability to analyse and address problem related to the design, planning, control, and improvements of manufacturing and service operations.
CO4	Understand the basic of Supply Chain Management.

(Note- Faculty can add the more CO's as per their subject.)**SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, word limit 100-150 words)**

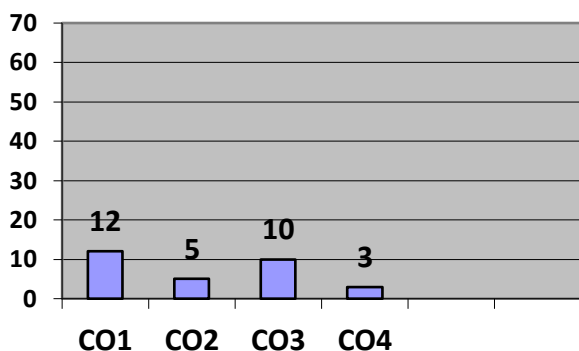
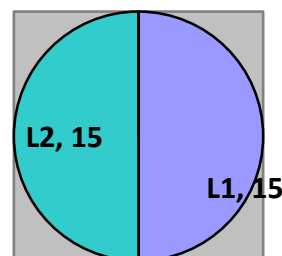
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	3	CO - 1	L 1
(B)	3	CO – 1	L1
(C)	3	CO – 1	L1
(D)	3	CO – 4	L1
(E)	3	CO - 1	L1

SECTION-B**Attempt Any two questions out of the following: (Long Answer Type Questions, word limit 200-250 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	5	CO – 3	L2
2 (B)	5	CO – 3	L2
3 (A)	5	CO – 3	L2
3 (B)	5	CO – 3	L2

SECTION-C**Attempt any one parts of the following: (Caselet / Numerical, , word limit 200-250 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4(A)	5	CO – 2	L2
4(B)	5	CO – 2	L2

Course Outcome Wise Marks Distribution**L1 L2 L3****Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2024-26)

(TRIMESTER –II) END TERM EXAMINATION, JANUARY - 2025

Operations Management

[Time: 1:00 Hours]

[Maximum Marks: 30]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(3×5=15)

- (a) How does Operations Management ensure efficient utilization of resources in an organization?
- (b) What does Strategic Operations Management mean?
- (c) Describe the primary benefits of implementing a Flexible Manufacturing System in a production facility.
- (d) Define the key components of a supply chain.
- (e) What is a transformation process in Operations Management? Explain with examples.

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(5×2=10)

- (a) Consider a company that manufactures both customized high-end furniture and mass-produced home furniture. The high-end furniture is made to order with unique designs, while the mass-produced furniture is standardized and produced in large quantities.

Given the differences in product characteristics (volume and variety), how should the company choose between a job shop, batch, mass, or continuous process for each product type? What factors should influence the decision, and how could the company's process choice impact efficiency, cost, and flexibility?

OR

- (b) What objectives should the company prioritize in its facility location decision? Which factors should influence the company's choice of location? How might the decision impact the company's long-term competitiveness and operational efficiency?

- Q.3. (a)** A company produces a wide range of products, some of which are high-volume and standardized, while others are low-volume and custom-made. What are the main characteristics of different types of layouts and how should the company select the most suitable layout type based on its product mix and production requirements?

OR

- (b) A company that manufactures both seasonal and non-seasonal products needs to balance its production capacity with fluctuating demand. How should the company approach aggregate planning to meet these demands, and how can master production scheduling (MPS) be used to ensure smooth operations and avoid stock outs or excess inventory?

SECTION-C

Q.4. Attempt all parts of the following:

(1×5=5)

AutoParts Co., a mid-sized automotive parts manufacturer, has been in business for over 15 years, supplying parts to several major automotive brands. Recently, the company began to face significant inefficiencies in its production processes. The manufacturing plant was experiencing high levels of waste, excessive inventory, and slow turnaround times, which resulted in higher costs and delays in meeting customer demand. Despite efforts to address these issues through traditional process improvement methods, AutoParts Co. still found it challenging to improve overall productivity.

In response, the company's leadership decided to adopt lean production principles, a methodology designed to enhance productivity by eliminating waste, improving processes, and creating more value for customers with fewer resources. The first steps in the lean transformation included intensive employee training on the identification and elimination of waste, as well as the introduction of the Kanban system for better inventory control. Additionally, the production processes were restructured to streamline operations and reduce bottlenecks.

However, the shift to lean production faced challenges from the outset. Many long-time employees were resistant to change, particularly those who had grown accustomed to traditional ways of working. The new lean practices required a cultural shift and greater collaboration between departments, which proved difficult to achieve. Despite these obstacles, some improvements were observed: lead times were reduced by 15%, and the company was able to cut down on inventory by 20% within the first six months.

Yet, AutoParts Co. still struggled with fully embedding lean practices across the organization. While some employees had embraced the changes, others continued to resist, resulting in inconsistent implementation of lean principles across different production lines. Management recognized that for lean production to be truly effective, the entire organization needed to be aligned and fully committed to the continuous improvement mindset.

4(a) What types of waste could AutoParts Co. aim to eliminate through lean production. Provide examples specific to their industry.

OR

4(b) Explain how the Kanban system works and how it could benefit AutoParts Co. in controlling inventory. What challenges might the company face in fully adopting lean production, and how could they overcome these challenges?